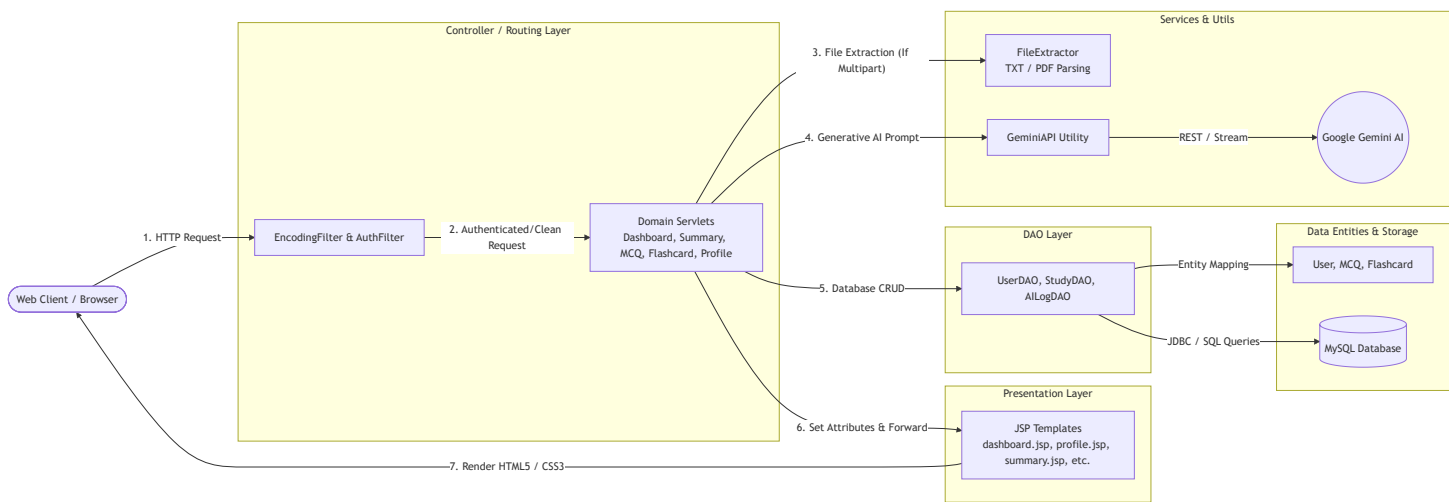


Architecture & Flow Diagrams

This document illustrates the complete software architecture of the **AI Study Companion** application. It details the core **Model-View-Controller (MVC)** design pattern implementation, full static class structural relationships, and dynamic interaction sequences across all major user pathways.

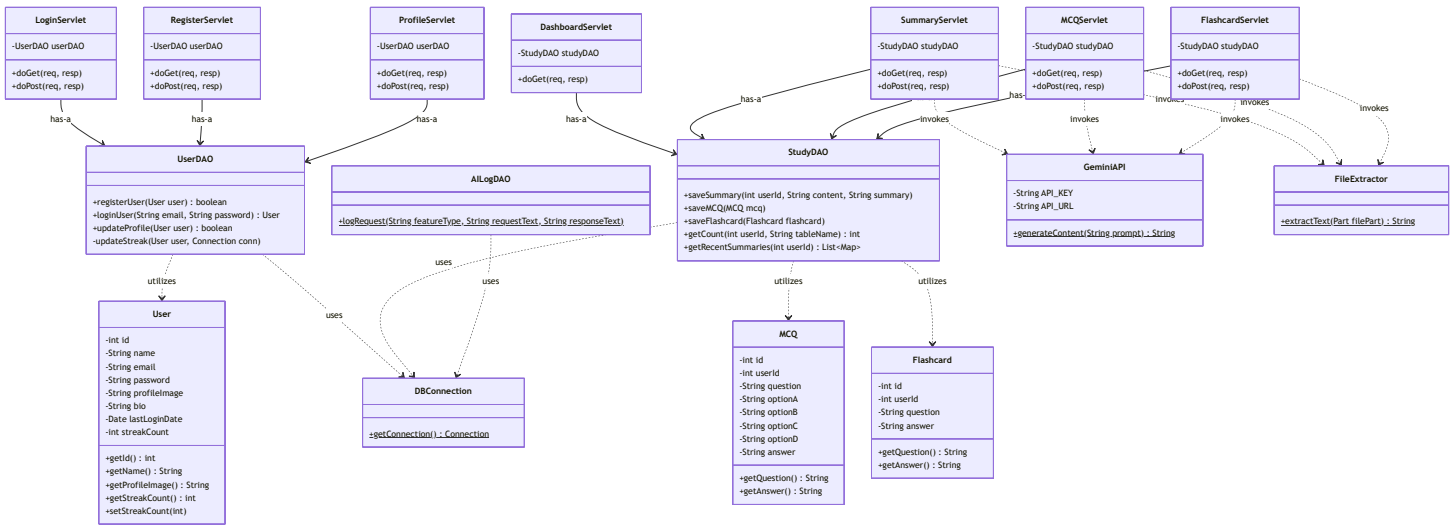
1. High-Level MVC Design Pattern Flow

The application follows the classic Java EE **MVC (Model-View-Controller)** pattern where HTTP requests pass through standard lifecycle filters before reaching domain-specific Servlets (Controllers). The Servlets utilize Data Access Objects (DAOs) to interact with the relational MySQL Database (Models) and forward processed structures to JavaServer Pages (Views) for presentation.



2. Complete Class Diagram

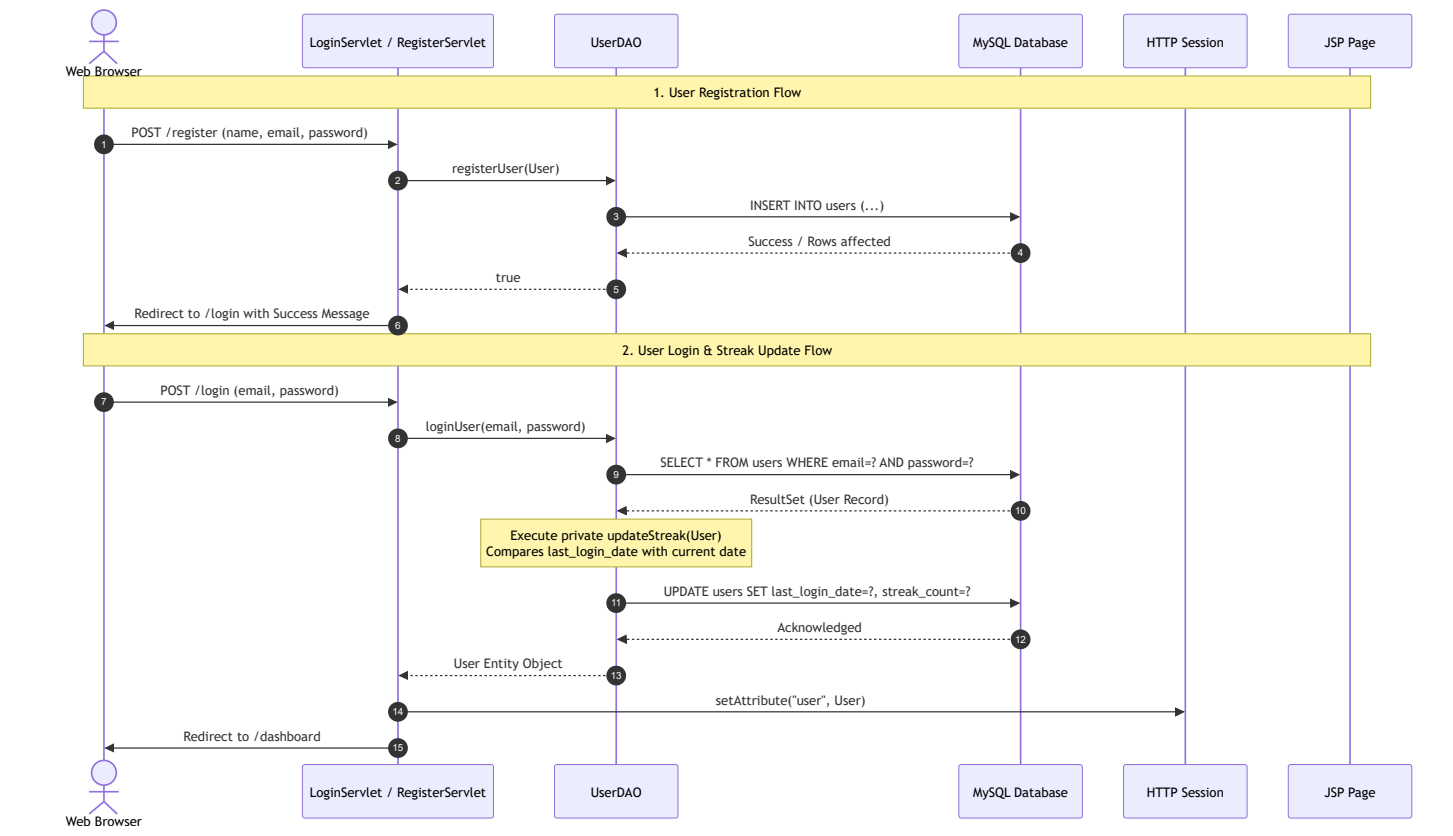
Below is the structured representation of all classes present within the project packages (`com.aistudy.*`). It showcases inheritance, field compositions, and service calls.



3. Sequence Diagrams for Core Application Flows

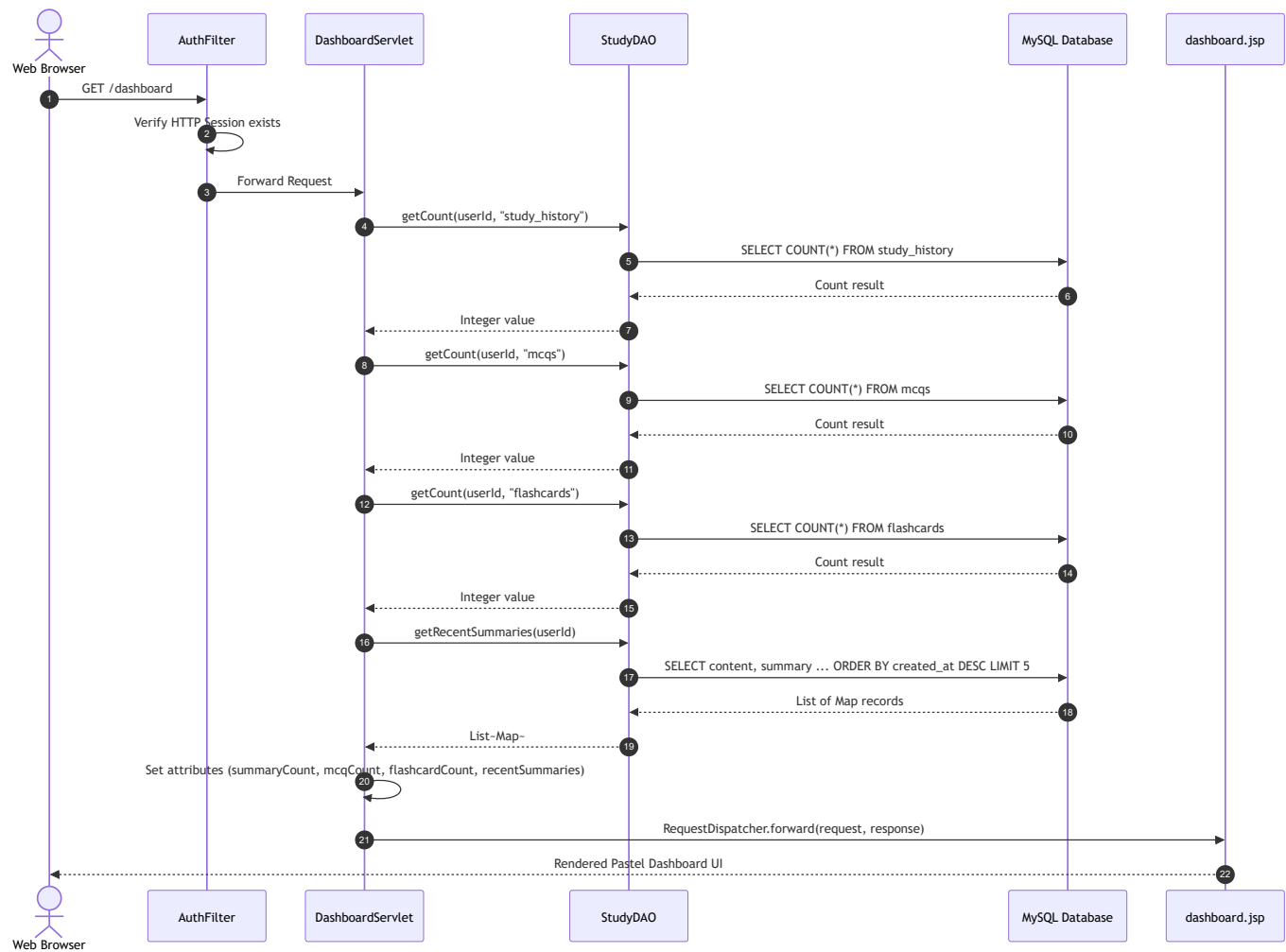
3.1 Authentication Flow (Registration & Login)

Illustrates how users onboard, authenticate, and update consecutive study streaks.



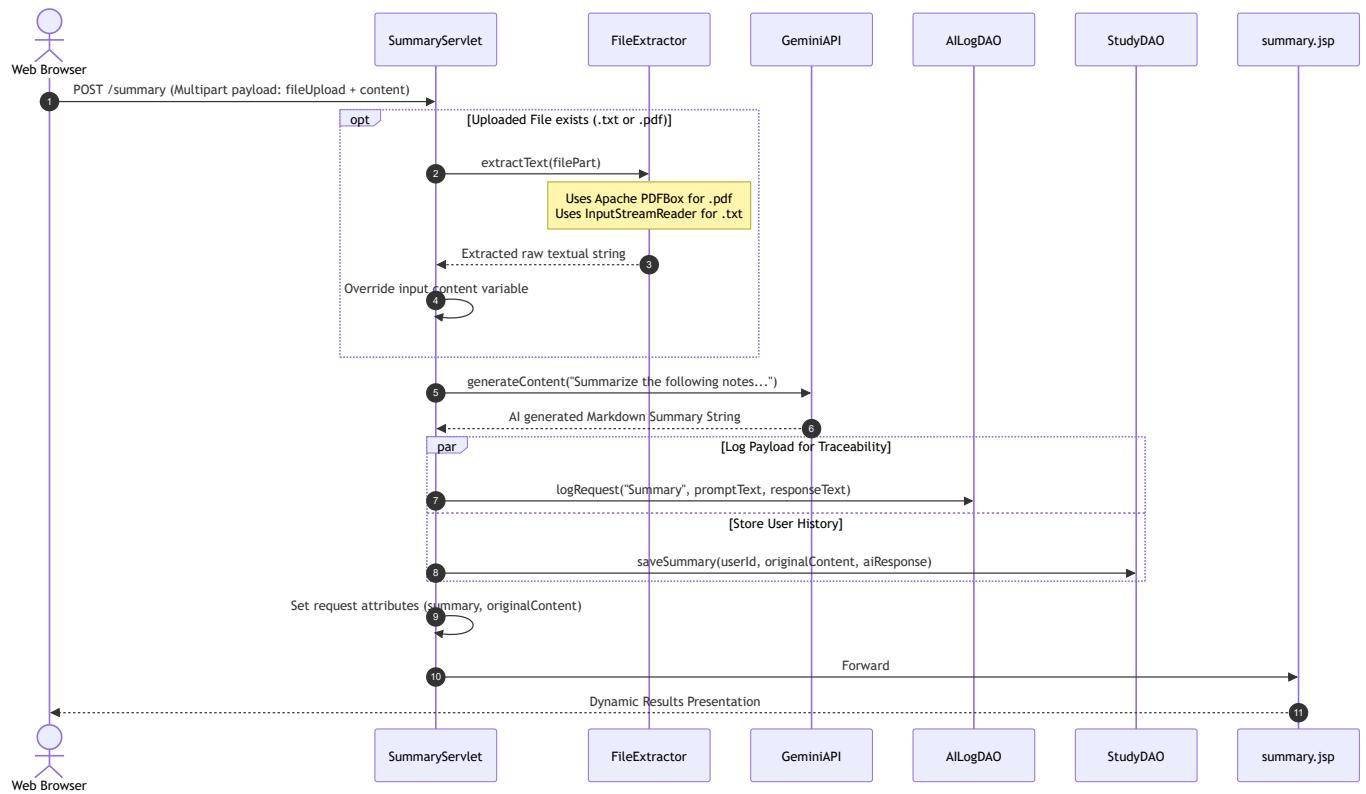
3.2 Dashboard Loading Flow

Highlights how user history and statistical counts are aggregated dynamically upon loading the landing interface.



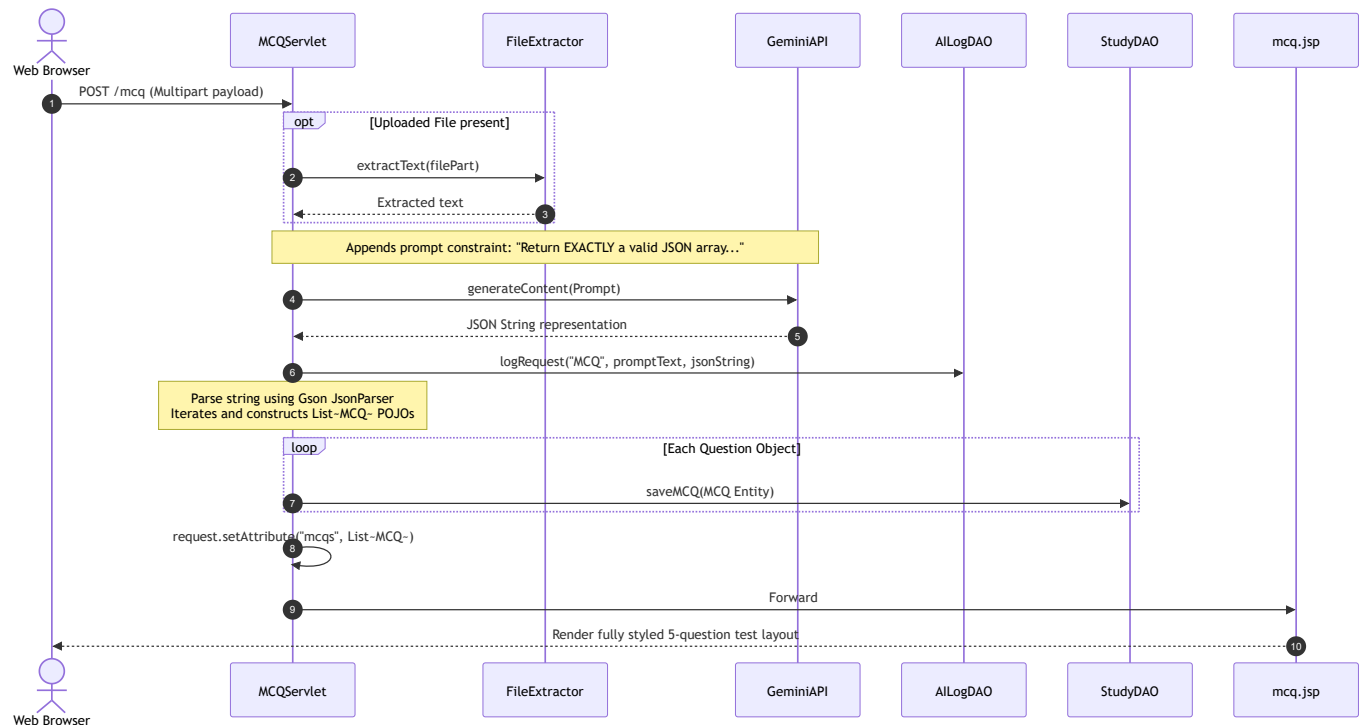
3.3 AI Summary Generation Flow (With File Upload Integration)

Demonstrates multi-format file extraction, external API invoking, and persistent payload logging.



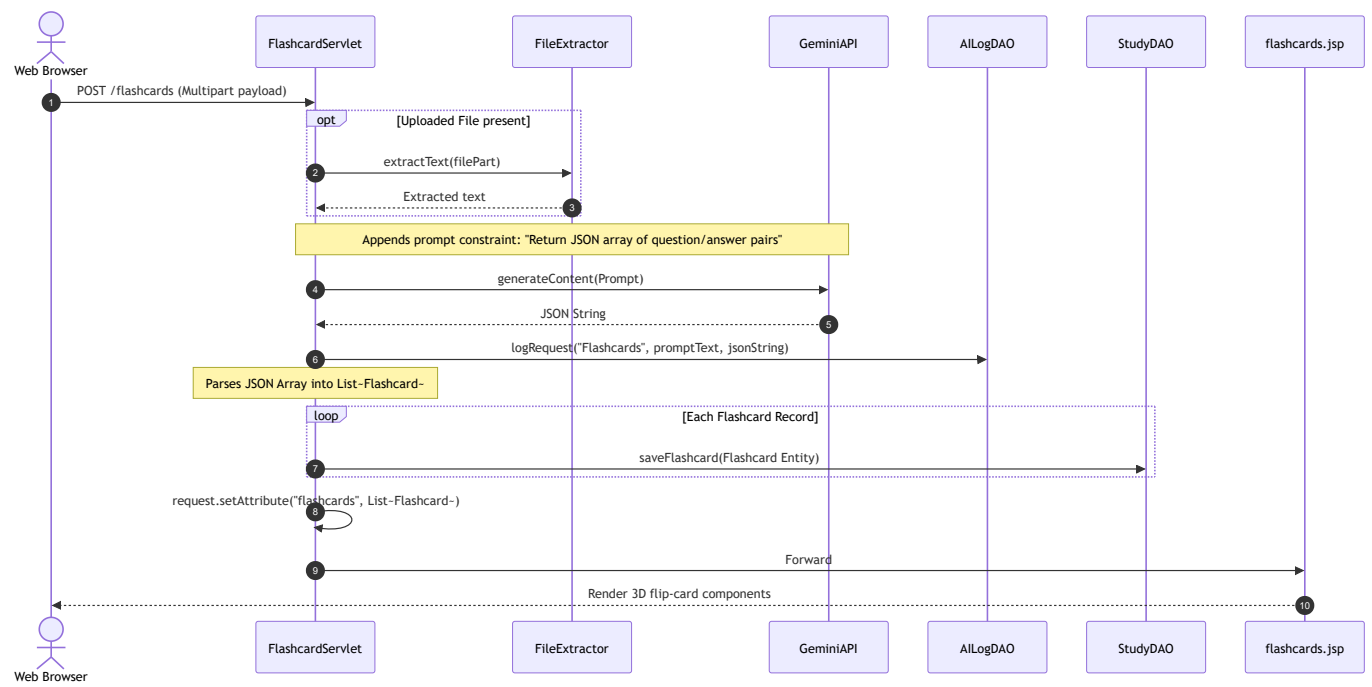
3.4 AI MCQ Generation Flow

Demonstrates JSON structured responses mapped into domain POJOs.



3.5 AI Flashcard Generation Flow

Demonstrates extracting Q&A structures mapped to reactive CSS components.



3.6 User Profile Updating & Avatar Upload Flow

Traces uploading multipart image assets natively to the server context directory.

